

# Oric Screen Editor

Screen editor for the Oric Atmos

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## Version history and download

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Link to latest builds: - .DSK disk image - .HFE disk image

Version v099-20220824-1345: - File picker added - Minor other tweaks and fixes

Version v099-20220615-1454: - First released beta version based on VDCSE  
version v099-20220324-1527



Figure 1: OSE Title Screen

## Introduction

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Oric Screen Editor is an editor to create text based screens for the Oric Atmos. It fully supports using user defined character sets.

**OSE-LOCI note:** this rewrite requires a **LOCI mass-storage device** to be attached to even start – the canvas itself is stored in LOCI-backed overlay RAM, not just file loading/saving. If no LOCI device is detected, the program shows a message and exits instead of starting the editor.

Main features of the program:

- Support for screen maps larger than 40x25 characters. Screens can be up to 10 KiB (10,240 bytes), all sizes fitting in that memory with width of 40 at minimum and height of 27 at minimum are supported.
- Supports resizing canvas size, clear or fill the canvas
- Support for loading user defined charsets (should be standard charsets of 96 characters of 6 bits width and 8 bits height, alternate charsets of 80 characters or combined charsets of 176 characters).
- Includes a simple character editor to change characters on the fly and directly see the result in your designed screen.
- Supports the Oric serial attributes for ink, paper and character modifiers (standard vs alternate charset, double height, blink).
- Write mode to freely type characters with the keyboard
- Line and box mode for drawing lines and boxes
- Select mode to cut, copy, delete

or repaint (only color or all attributes) the selection. - Move mode to scroll the screen contents (due to memory constraints only for the 40x27 viewport)

- Palette mode, including visual charmap mode, to visually select characters and colors
- Favorite slots to quickly select 10 favorite characters

- **OSE-LOCI addition:** supports an IJK-compatible joystick (Raxiss IJK interface) as an alternative to the keyboard everywhere cursor keys and ENTER are used – no separate mode or setting needed, it is detected automatically at startup and simply works alongside the keyboard.

- **OSE-LOCI additions:** Try mode to preview a character before plotting it; Jump (**J**) and Home (**H**) to move the cursor and viewport straight to a typed or fixed coordinate; a unified Find/Replace (**F**) for screencodes and ink/paper colors across the whole canvas; a hollow-box and ellipse/circle option in Line and box mode; a Charset menu option to reset the standard character set from ROM in one step; and a hex-direct attribute entry shortcut in Write mode.

## Known issues

- Filepicker routine only is properly working with SEDORIC3 disks created by the TAP2DSK tool from OSDK. If you want to import screens or charsets from other disks, please copy them to a TAP2DSK created image first.

## Start program

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Mount the OSE disk image. Choose the .DSK or .HFE image depending on what is supported on the hardware or emulator you use (in general: .HFE for Cumana Reborn, .DSK for emulator).

The OSE executable will autostart on mounting the disk. If not, just type OSE<ENTER>.

Description of contents of the disk image:

| Filename | Extension | Description                                      |
|----------|-----------|--------------------------------------------------|
| OSE      | .COM      | Main executable                                  |
| OSEHS1   | .BIN      | Help screen for main mode                        |
| OSEHS2   | .BIN      | Help screen for character edit and palette modes |
| OSEHS3   | .BIN      | Help screen for select, move and line/box modes  |
| OSEHS4   | .BIN      | Help screen for write and color write modes      |
| OSETSC   | .BIN      | Title screen                                     |

(Fun fact: all screens have actually been created using OSE as editor)

Leave the title screen by pressing any key.

## Main mode

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After the title screen, the program starts in this mode. At start the screen shows this:

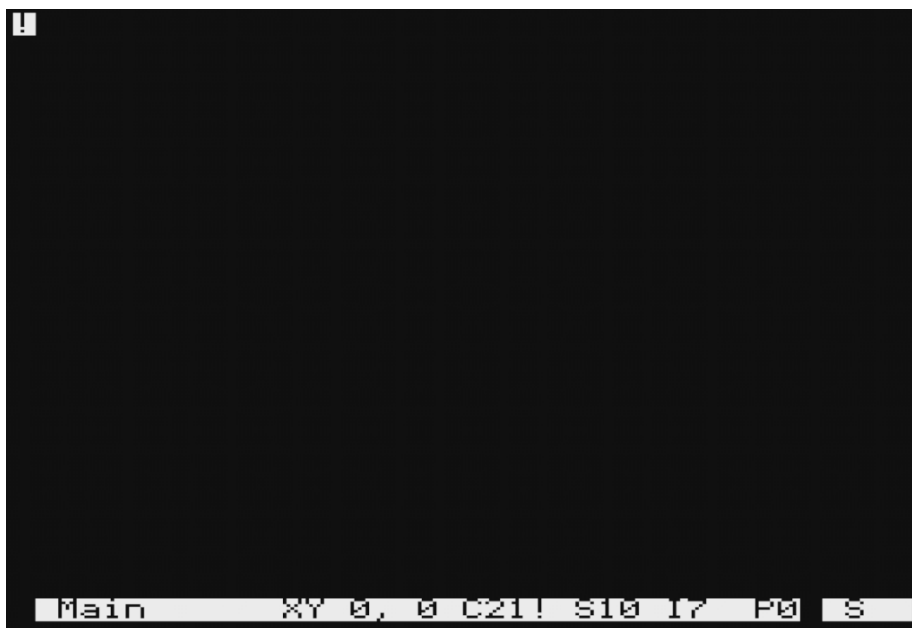


Figure 2: Screen in main mode

Only a inversed cursor with the presently selected screencode is visible.

Press these keys in main mode for editing:

| Key                | Description                                                   |
|--------------------|---------------------------------------------------------------|
| <b>Cursor keys</b> | Move cursor                                                   |
| <b>+ or =</b>      | Next character (increase screen code)                         |
| <b>-</b>           | Previous character (decrease screen code)                     |
| <b>0-9</b>         | Select character from favorite slot with corresponding number |
| <b>SHIFT + 0-9</b> | Store character to favorite slot with corresponding number    |
| <b>,</b>           | Previous ink (decrease color number)                          |
| <b>.</b>           | Next ink (increase color number)                              |
| <b>;</b>           | Decrease paper (decrease color number)                        |

| Key            | Description                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| '              | Increase paper (increase color number)                                                                               |
| <b>SPACE</b>   | Plot with present screen code and attributes                                                                         |
| <b>DEL</b>     | Clear present cursor position (plot white space)                                                                     |
| <b>B</b>       | Toggle ' <b>B</b> link' attribute                                                                                    |
| <b>A</b>       | Toggle <b>A</b> lternate or Standard character set                                                                   |
| <b>D</b>       | Toggle <b>D</b> ouble height attribute                                                                               |
| <b>I</b>       | Plot a change <b>I</b> nk modifier attribute for the present selected ink color                                      |
| <b>O</b>       | Plot a change <b>P</b> aper modifier attribute for the present selected paper color                                  |
| <b>U</b>       | Plot a character set modifier attribute for the present selected charset selection, double size and blink attributes |
| <b>E</b>       | Go to 'character <b>E</b> dit mode' with present screen code                                                         |
| <b>G</b>       | <b>G</b> rab underlying character and attribute at cursor position                                                   |
| <b>W</b>       | Go to ' <b>W</b> rite mode'                                                                                          |
| <b>L</b>       | Go to ' <b>L</b> ine and box mode'                                                                                   |
| <b>M</b>       | Go to ' <b>M</b> ove mode'                                                                                           |
| <b>S</b>       | Go to ' <b>S</b> elect mode'                                                                                         |
| <b>P</b>       | Go to ' <b>P</b> alette mode'                                                                                        |
| <b>C</b>       | Go to ' <b>C</b> olour picker'                                                                                       |
| <b>T</b>       | <b>T</b> ry mode                                                                                                     |
| <b>R</b>       | Toggle ' <b>R</b> everse': toggle increase/decrease screencode by 128                                                |
| <b>J</b>       | <b>OSE-LOCI new:</b> <b>J</b> ump (go to) a typed canvas X/Y coordinate                                              |
| <b>H</b>       | <b>OSE-LOCI new:</b> <b>H</b> ome – jump cursor and viewport to the canvas origin (0,0)                              |
| <b>F</b>       | <b>OSE-LOCI new:</b> <b>F</b> ind/Replace a screencode or ink/paper color across the whole canvas                    |
| <b>Z</b>       | <b>OSE-LOCI only:</b> Undo the most recent canvas edit                                                               |
| <b>Y</b>       | <b>OSE-LOCI only:</b> Redo the most recently undone edit                                                             |
| <b>FUNCT+1</b> | Go to main menu                                                                                                      |
| <b>FUNCT+6</b> | Toggle statusbar visibility                                                                                          |

| Key           | Description |
|---------------|-------------|
| <b>FUNT+8</b> | Help screen |

**NB:** Within Oricutron using the default keyboard mapping, the FUNCT key is assigned to the ALT key on the PC keyboard.

#### *Moving cursor*

Press the **cursor keys** to move the cursor around the screen. If the canvas size is bigger than the 40x27 screensize, the screen will scroll on reaching the edges.

#### *Selecting the screencode to plot*

The **+** or **-** key will increase resp. decrease the selected screencode by one. The cursor will update to the presently selected screencode.

Pressing **R** will increase the screencode by 128 if the present screencode is lower than 128, otherwise decrease by 128. This will enable/disable the reverse bit for the hardware reversing of the character.

#### *Selecting the screencode to plot from a favorite slot*

In OSE 10 positions are available to store your most frequently used characters in. Pressing one of the **0-9** keys selects the favorite with the corresponding number.

#### *Storing the present screencode to a favorite slot*

Pressing **SHIFT** plus **0-9** stores the presently selected character to the corresponding favorite slot.

#### *Selecting the attributes to plot*

Increase or decrease the color code by one by pressing the **.** resp. **,** key for the Ink color, or **;** resp. **'** for the Paper color. Pressing **B**, **D** or **A** will toggle the **B**link, **D**ouble size or **A**lternate charset attribute. Alternatively, press **C** to open the colour picker and select the Ink and Paper colors together from a visual grid.

#### *Plotting and deleting a character*

Press **SPACE** to plot the presently selected character at the present cursor position. **DEL** will delete the character or attribute value at the present position and replace it with a SPACE as character. Press **T** to preview how the selected character would look like if plotted without cursor blinking. Then press **SPACE** to confirm plotting the character, or any other key to cancel.

#### *Plotting serial attributes*

Due to the way the Oric handles color and attribute changes, every screen position is either an attribute modifier or a character, but never both at the same time. For this reason, on plotting a character no attribute is plotted. The

user of OSE needs to plot the serial attributes separately by its own desire. To do so, press **I** for plotting an **I**nk modifier for the present ink color, **P** for plotting a **P**aper modifier for the present paper color and **U** for plotting a character set modifier for the present character attributes (the alternate, double and blink toggles).

#### *Grabbing a character*

Pressing **G** will ‘grab’ the character or attributes at the present cursor position and change the selected character screencode or attribute to these values for use in all other edit functions.

#### *Character edit mode*

This will enter character edit mode and start with editing the presently selected screencode. Tip: if you want to edit a specific character on the screen, grab that character first by moving the cursor on that character and press **G** for grab.

#### *Enter edit modes*

Press **S** (Select mode) , **M** (Move mode), **L** (Line and box mode) or **W** (Write mode) for entering the corresponding edit modes. Reference is made to the specific sections in this readme for these modes (click the links). From all modes, return to main mode by pressing **ESC**.

#### *Jump to coordinates and Home (OSE-LOCI new)*

Press **J** to open a popup asking for an X then a Y canvas coordinate (both pre-filled with the cursor’s current position). Confirming both jumps the cursor and viewport there in one step, scrolling as needed – much quicker than scrolling cell by cell on a canvas bigger than the 40x27 screen. ESC at either field cancels with no change. Press **H** to jump straight back to the canvas origin (0,0) without any popup.

#### *Find/Replace (OSE-LOCI new)*

Press **F** to open the Find/Replace popup. First choose what to search for: **1** for screencode, **2** for ink color, or **3** for paper color. Next, enter the value to find (a hexadecimal screencode, or a color number 0-7). Finally, either press **ENTER** with a replacement value to replace every matching occurrence across the whole canvas (this can be undone afterwards with **Z**), or press **ESC** at that step to instead jump the cursor straight to the next occurrence of the value you searched for, without changing the canvas. ESC at the first two steps cancels the whole operation.

#### *Undo and redo (OSE-LOCI only)*

Press **Z** to undo the most recent canvas edit (plotting, Line/Box, Select fills/cut/copy, Move, Write mode, or Screen > Clear/Fill), and **Y** to redo the most recently undone edit. This needs a LOCI device attached (undo history is stored in the Oric’s overlay RAM, which only a LOCI device can enable) — without one, **Z/Y** simply do nothing. There is no V1 equivalent of this feature.

*Toggle statusbar visibility* Press **FUNCT+6** to toggle between the statusbar being visible (default) or not.

*Help screen* Press **FUNCT+8** to show a help screen with all keyboard commands for this mode.

## Statusbar

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If enabled, the statusbar is plotted as this at the lowest line of the screen:



Figure 3: Status bar

From left to right, this status bar shows:

- Mode: mode the program is in (such as Main, Select, Line/Box, Palette or Character Editor).
- X,Y: X and Y coordinates of the cursor (coordinates of the large full screen, and not only the visible screen, if a larger screen than 40 by 27 characters is selected. Normally shows in decimal, but shows in hexadecimal if X or Y maximum size is higher than 99)
- C: Screencode, the present selected character to plot, first as screencode number in hexadecimal, second as actual visual character.
- S: The screencode in memory underneath the cursor position. Will show either the character code (bigger than \$20) or the attribute code (smaller than \$20) in hexadecimal
- I: Ink, the present selected color for the ink. First as number 0-7, then as visual color.
- P: Paper, the present selected color for the paper. First as number 0-7, then as visual color.
- The last three positions show the character set modifier attributes that are enabled: A if the alternate charset is enabled, D for double size and B for blink.

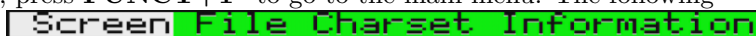
The status bar auto hides if the cursor is moved to the lowest visible line on the screen, and pops up again (if enabled in the first place) when the cursor moves up.

Pressing **FUNCT+6** toggles statusbar visibility in every mode.

## Main menu

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From main mode, press **FUNCT+1\*** to go to the main menu. The following menu will pop up:





(NB: Note that if your design uses a changed character set, the program will load the standard system font and your design might temporarily look incorrect. This will be restored on exiting the main menu. Also, the pulldown menus will show a black shade to the right due to Oric system limitations).

Navigation in this menu is performed by the following keys:

| Key                        | Description                        |
|----------------------------|------------------------------------|
| <b>Cursor LEFT / RIGHT</b> | Move between main menu options     |
| <b>Cursor UP/ DOWN</b>     | Move between pulldown menu options |
| <b>RETURN</b>              | Select highlighted menu option     |
| <b>ESC</b>                 | Leave menu and go back             |

### *Screen menu*



Figure 4: Screen menu

### *Width: Resize width*

Resize the canvas width by entering the new width. You can both shrink as expand the width. Minimum width is 40, maximum width depends on the canvas height and the result fitting in the maximum of 10 KiB memory size allocation.

Note that with shrinking the width you might loose data, as all characters right of the new width will be lost. That is why on shrinking a pulldown menu will pop-up asking if you are sure. Select the desired answer (yellow highlighted position if using a black background).

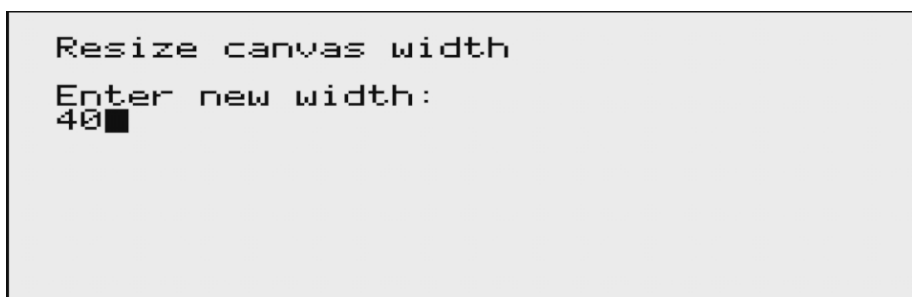


Figure 5: Resize width

### *Height: Resize height*

Similar to resize width, with this option you can resize the height in the same way. Minimum height is 27, maximum again dependent on width given maximum of 10 KiB memory allocation.

Also here: on shrinking you might loose data, which is lost if you confirm.

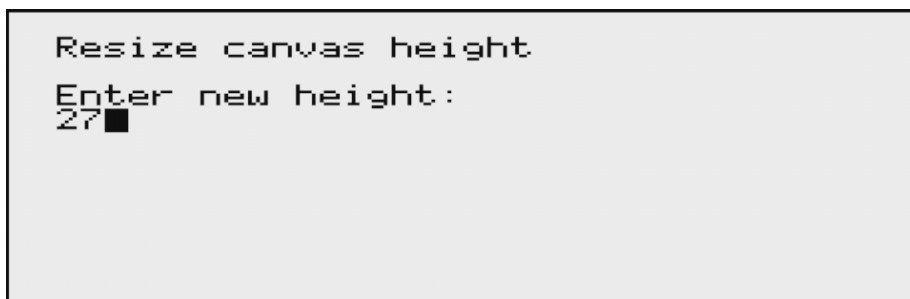


Figure 6: Resize height

*Clear: Clear the canvas*

Selecting this menu option will clear the canvas (which means filling the canvas with spaces, with attribute code for the selected color and selected paper plotted in the first two columns of the screen). No confirmation will be asked.

*Fill: Fill the canvas*

Similar to clear, but this will fill the canvas with the present selected screencode (so the values that the cursor was showing).

**File menu**

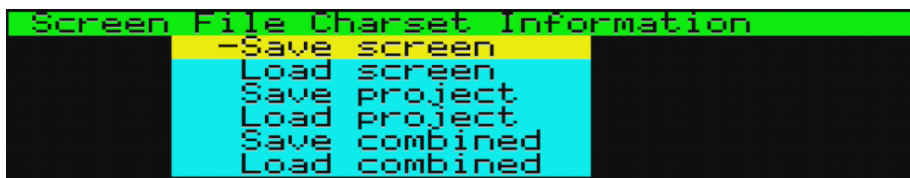


Figure 7: File menu

**OSE-LOCI note:** this rewrite uses the **LOCI mass-storage device** for all file I/O instead of V1's tape commands. **Save** actions ask for a typed filename (up to 48 characters). **Load** actions open a file picker instead: a scrollable directory listing (cursor keys to move, ENTER to descend into a folder or select a highlighted file, LEFT to go back up to the parent folder, ESC to cancel) showing only the files relevant to the action you chose (e.g. Load Project only shows project files). If no LOCI device is detected, every File/Charset menu item shows a "No LOCI device detected" message instead of attempting the operation, so the rest of the editor keeps working normally without one attached.

### *Save screen / Load screen*

Saves or loads just the canvas (no character sets) to/from <filename>.BIN on the LOCI device: a small header recording the canvas width/height, followed by the screen data. Loading applies the saved width/height automatically – no need to enter them separately.

### *Save project / Load project*

Saves or loads the canvas together with its metadata (cursor position, viewport, ink/paper/blink/double/altchar selection) and, if you’ve edited them this session, both character sets – as up to four files sharing your typed filename: <filename>PJ.BIN (metadata), <filename>SC.BIN (screen), <filename>CS.BIN (standard charset, only written/expected if you edited it) and <filename>CA.BIN (alternate charset, same condition). Loading a project that was saved with only one charset edited leaves the other charset bank untouched (still whatever was loaded/default beforehand).

### *Save combined / Load combined*

Saves or loads the canvas together with the standard character set in a single <filename>.BIN file (header + the standard charset’s 768 bytes + the screen data).

### *Charset: Load and save character set*

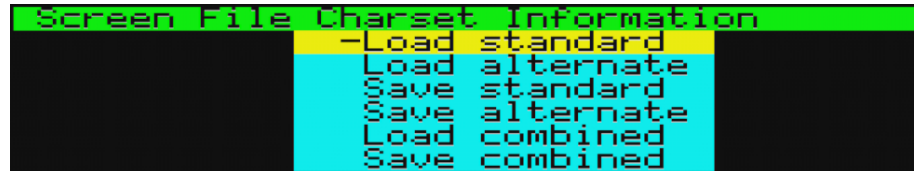


Figure 8: Charset menu

Load or save the standard or alternate character set separately (768 bytes each, <filename>.BIN), or “combined”: saving combined is identical to saving the standard set; loading combined loads the file into **both** the standard and alternate banks, so they end up identical. (This differs from V1, which used a ROM call to regenerate the alternate set from the standard one on load – that ROM call doesn’t work in this rewrite, so copying the same data into both banks is the closest available equivalent.)

### *Reset Std->ROM (OSE-LOCI new)*

Restores the standard character set from the Oric’s ROM font in one step, discarding any edits made to it via the character editor this session. Asks for confirmation first, since this cannot be undone (use this if you want to start a fresh standard character set without re-entering the editor and pressing **S** glyph

by glyph). Only the standard set can be reset this way – the Oric’s ROM has no alternate character set table to restore from.

*Information: Version information, exit program*



Figure 9: Information menu

*Information*

**OSE-LOCI note:** this option shows a 2-page popup instead of V1’s single text screen, identical to the locifilemanager-v2 LOCI file manager’s own version screen: a page with the IDreamtIn8Bits.com logo plus version and credits text, followed by a page with a QR code linking to this project’s GitHub page. Press any key to advance through the pages and to return to the main menu afterwards.

*Exit program*

**OSE-LOCI note:** since this rewrite’s bare-metal runtime has no underlying tape-loader process to return to (unlike the original CC65-based OricScreenEditor), this option resets the machine back to its cold-start state instead. NB: No confirmation will be asked and unsaved work will be lost.

## Character editor:

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Pressing **E** from the main mode will result in the character editor popping up, which looks like this:

It shows a magnified grid of the bits in the present character. The header shows the screencode of the present character (in hex) and if the Standard (Std) or Alternate (Alt) character set is active. On the left of the grid the byte values of the 8 lines of the character are shown in hex.

Keyboard commands in this mode:

| Key                | Description                                                   |
|--------------------|---------------------------------------------------------------|
| <b>Cursor keys</b> | Move cursor                                                   |
| <b>+</b>           | Next character (increase screen code)                         |
| <b>-</b>           | Previous character (decrease screen code)                     |
| <b>0-9</b>         | Select character from favorite slot with corresponding number |
| <b>SHIFT + 0-9</b> | Store character to favorite slot with corresponding number    |

| Key                  | Description                                                                  |
|----------------------|------------------------------------------------------------------------------|
| <b>SPACE</b>         | Toggle pixel at cursor position (plot/delete pixel)                          |
| <b>DEL</b>           | Clear character (delete all pixels of present character)                     |
| <b>I</b>             | Inverse character                                                            |
| <b>Z</b>             | Undo: revert present character to original state                             |
| <b>S</b>             | Restore character from system character set (=lower case system ROM charset) |
| <b>C</b>             | Copy present character                                                       |
| <b>V</b>             | Paste present character                                                      |
| <b>X / Y</b>         | Mirror in <b>X</b> axis or <b>Y</b> -axis                                    |
| <b>L / R / U / D</b> | Scroll <b>Left</b> , <b>Right</b> , <b>Up</b> or <b>Down</b>                 |
| <b>H</b>             | Input <b>Hex</b> value for line at cursor position                           |
| <b>ESC</b>           | Leave character mode and go back to main mode                                |
| <b>FUNCT+6</b>       | Toggle statusbar visibility                                                  |
| <b>FUNCT+8</b>       | Help screen                                                                  |

#### *Moving cursor*

Press the **cursor keys** to move the cursor around the 8 by 8 grid.

#### *Selecting the screencode to plot*

The **+** or **-** key will increase resp. decrease the selected screencode by one. Pressing **A** will toggle the character set to be used between Standard and Alternate.

#### *Selecting the screencode to plot from a favorite slot*

In OSE 10 positions are available to store your most frequently used characters in. Pressing one of the **0-9** keys selects the favorite with the corresponding number.

#### *Storing the present screencode to a favorite slot*

Pressing **SHIFT** plus **0-9** stores the presently selected character to the corresponding favorite slot.

#### *Toggling bits in the grid*

Press **SPACE** to toggle the bit at the present cursor position. **DEL** clears all bits in the grid, **I** inverts all bits in the grid.

#### *Undo and restore*

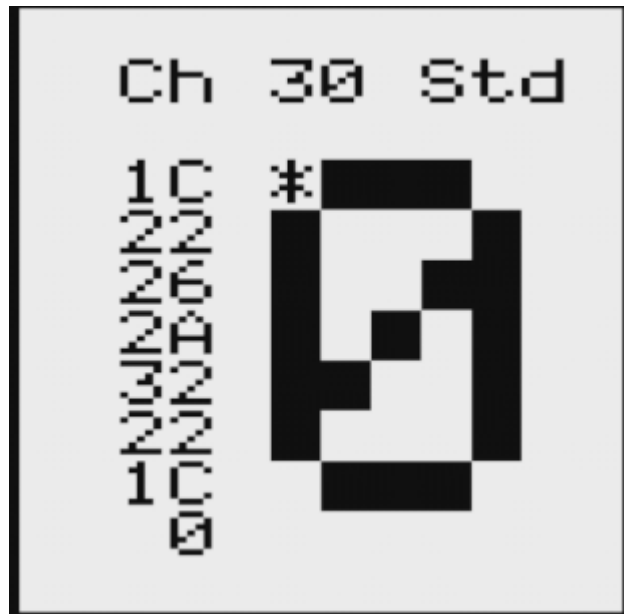


Figure 10: Character editor

**U** reverts the present character to the original values. Note that after changing to a different screncode to edit, the previous screncode can no longer be reverted.

**S** copies the present screncode from the system font.

#### *Copy and paste*

**C** copies the present screncode to buffer memory to be pasted with pressing **V** at a different screncode after selecting this other screncode.

#### *Mirror and scroll*

Press **X** or **Y** to mirror the grid on the X resp. Y axis. **L**, **R**, **U** and **D** scrolls the grid to the **Left**, **Right**, **Up** or **Down**.

#### *Hex input*

Press **H** to edit the full present row of the grid by entering the hex value of the byte representing the bits in that byte for the line.

#### *Leave mode and help*

Pressing **ESC** leaves the character mode and returns to main mode. **FUNCT+8** will show a help screen with all keyboard commands of the character mode.

## Palette mode:

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Pressing **P** in the main mode starts the Palette mode. In this mode a character for plotting can be selected from the character set, the full 121 color palette and the 10 favorite slots.

A window like this appears:

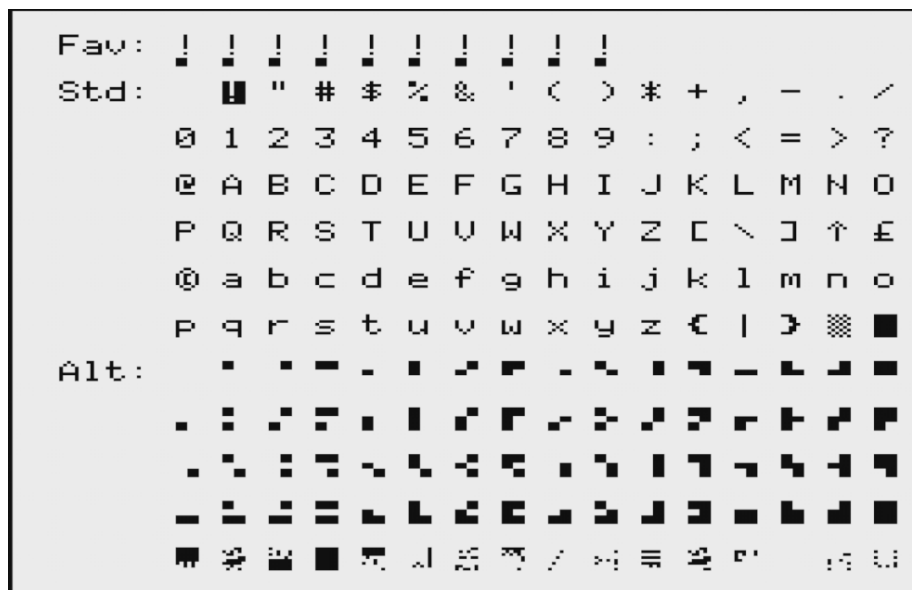


Figure 11: Palette mode screenshot

The window shows the 10 favorite slots as first line, below that the standard character set and below that the alternate character set.

Keyboard commands in this mode:

| Key                   | Description                                        |
|-----------------------|----------------------------------------------------|
| <b>Cursor keys</b>    | Move cursor                                        |
| <b>SPACE or ENTER</b> | Select character                                   |
| <b>0-9</b>            | Store character in corresponding favorite slot     |
| <b>V</b>              | Toggle between normal mode and visual charmap mode |
| <b>ESC</b>            | Leave character mode and go back to main mode      |
| <b>FUNCT+6</b>        | Toggle statusbar visibility                        |

*Moving cursor*

Press the **cursor keys** to move the cursor around the grid. You can move over to the different sections by just moving out of a section to the other.

#### *Selecting character or color*

Press **SPACE** or **ENTER** to select the highlighted character or color as new character or color to plot with. This leaves the palette mode.

#### *Storing to a favorite slot*

Pressing **0-9** stores the presently highlighted character to the corresponding favorite slot.

#### *Toggle visual charmap mode*

Visual charmap mode is a mode in which the palette for the alternate charset is mapped in such a way that characters are ordered in a logical order for drawing. This mode makes only sense for unchanged alternate charsets.

This looks like this:

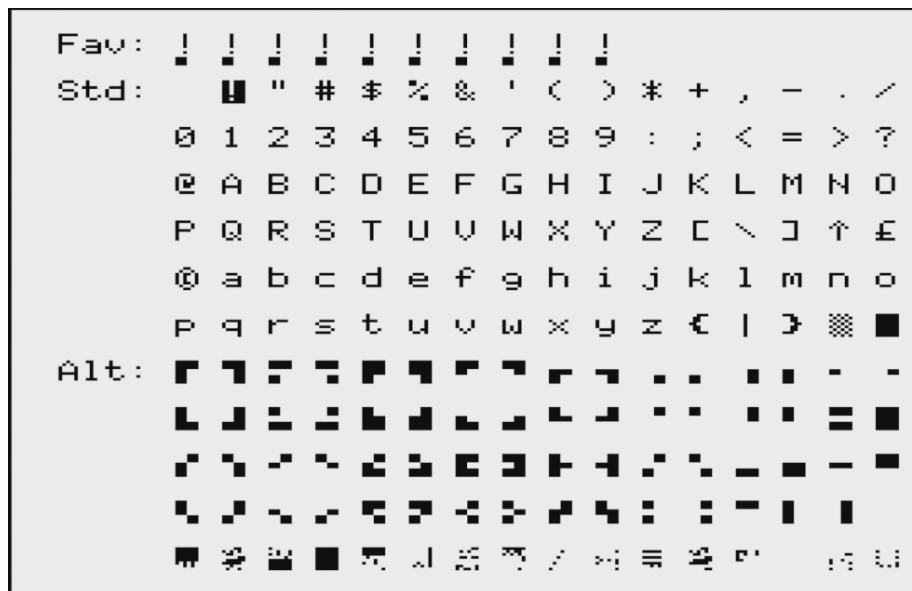


Figure 12: Visual PETSCII palette

Pressing **V** toggles between normal and visual mode.

#### *Leave mode and help*

Pressing **ESC** leaves the palette mode and returns to main mode. There is no separate help screen for the palette mode.



## Colour picker

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Pressing **C** in main mode starts the Colour picker. This is an addition over the original OricScreenEditor: a visual grid for selecting the Ink and Paper colors to plot with together, including a preview of the resulting normal and inverse color combination.

A window like this appears, showing an 8x8 grid of all Ink/Paper combinations, plus feedback lines for the currently highlighted Ink, Paper and the resulting normal/inverse preview:

Select ink and paper colour

<8x8 grid of ink x paper colour cells, each showing the normal and inverse swatch for that c

Ink: N <swatch>

Paper: N <swatch>

Result: <normal/inverse preview>

Keyboard commands in this mode:

| Key                   | Description                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------|
| <b>Cursor keys</b>    | Move cursor (LEFT/RIGHT cycle Ink, UP/DOWN cycle Paper, both wrap 0-7)                            |
| <b>SPACE or ENTER</b> | Select the highlighted Ink/Paper combination                                                      |
| <b>ESC</b>            | Leave the colour picker and go back to main mode, leaving the selected Ink/Paper colors unchanged |

### *Moving cursor*

Press the **cursor keys** to move the cursor around the 8x8 grid. **LEFT/RIGHT** cycle the Ink color, **UP/DOWN** cycle the Paper color, both wrapping between color 0 and 7.

### *Selecting ink and paper*

Press **SPACE** or **ENTER** to select the highlighted Ink/Paper combination as the new colors to plot with. This leaves the colour picker and returns to main mode; the statusbar's I and P fields update accordingly.

### *Leave mode*

Pressing **ESC** leaves the colour picker without changing the selected Ink/Paper colors, and returns to main mode.

## Select mode:

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Pressing **S** in the main mode starts the Select mode.

If enabled, the statusbar shows this on entering this mode:



Figure 13: Status bar in Select mode

In this mode a selection can be made on which different operations can be performed as described below.

| Key                | Description                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>X</b>           | Cut: Delete selection at old position and paste at new position                                                      |
| <b>C</b>           | <b>C</b> opy: Copy selection at new position, leaving selection unchanged at old position                            |
| <b>D</b>           | <b>D</b> elete selection (fill with spaces)                                                                          |
| <b>I</b>           | Paint with <b>I</b> nk attribute: plot ink modifier in presently selected color                                      |
| <b>P</b>           | Paint with <b>P</b> aper attribute: plot paper modifier in presently selected color                                  |
| <b>M</b>           | Paint with Character <b>M</b> odifier attribute: plot character modifier in presently selected A,D and B attributes. |
| <b>RETURN</b>      | Accept selection / accept new position                                                                               |
| <b>ESC</b>         | Cancel and go back to main mode                                                                                      |
| <b>Cursor keys</b> | Expand/shrink in the selected direction / Move cursor to select destination position                                 |
| <b>FUNCT+6</b>     | Toggle statusbar visibility                                                                                          |
| <b>FUNCT+8</b>     | Help screen                                                                                                          |

### *Making the selection*

Ensure that the cursor is located at a corner of the selection to be made before entering Select mode. On entering select mode, grow the selection by pressing the **Cursor keys** to increase or decrease width and height in the desired direction from the origin. This is similar to the keys used in the Line and box mode.

The selection will be visually shown with plotting in the present selected screen-code and attributes. It should look like this:



Figure 14: Select mode

Accept the selection by pressing **RETURN**, cancel the selection by pressing **ESC**.

*Choose action to perform*

After accepting the selection, press **X**, **C**, **D**, **A** or **P** to choose an action, or press **ESC** to cancel. Statusbar (if enabled) shows this as prompter:



Figure 15: Statusbar Select Options

*Cut and copy*

After pressing **X** for cut or **C** for copy, move cursor to the upper left corner where the selection should be copied to. **C** will only make a copy, **X** will delete the selection at the old location.

Statusbar (if enabled) displays Cut or Copy correspondingly, like:

*Delete*

Pressing **D** will erase the present selection (fill the selected area with spaces).



Figure 16: Statusbar Cut or Copy

*Paint with attribute or only color*

Pressing **I**, **P** or **M** will fill the area with resp. ink color, paper color or character set modifiers with the presently selected values. Makes most sense in selecting a vertical length of one character wide to change attributes for all lines right of that line.

*Leaving mode and Help*

Leave selection mode by pressing **ESC**. Pressing **FUNCT+8** at any time in this mode will provide a helpscreen with the key commands for this mode (not possible if the selection is grown but not yet accepted).

**OSE note:** Cut (**X**) and Copy (**C**) move the selection to a new position you pick with the cursor keys (**ENTER** to confirm, **ESC** to cancel) — Copy leaves the original in place, Cut clears it. If the destination would extend past the edge of the canvas, a “Selection does not fit.” message appears and nothing is changed.

## Move mode:

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Pressing **M** in the main mode starts the Move mode. Use this mode to scroll the present viewport in the desired direction by pressing the **Cursor keys**.

Note that moving is only performed on the present 40x25 viewable part of the screen, so on larger canvas sizes not the whole screen will be moved. This is due to memory constraints.

It is also important to note that characters that ‘fall off’ of the screen are lost if the move is accepted.

Alternative to move mode is using select mode and use Cut to move a selection to a new position.

Accept with **RETURN**, cancel with **ESC**. Both will leave this mode and return to main mode.

**OSE note:** in this LOCI-based rewrite, each move is applied directly as you press a cursor key (there’s no separate scratch copy to roll back from) — so **RETURN** and **ESC** behave identically here, both keeping whatever shifts you’ve already made this session.

| Key                | Description                     |
|--------------------|---------------------------------|
| <b>Cursor keys</b> | Move in the selected direction  |
| <b>RETURN</b>      | Accept moved position           |
| <b>ESC</b>         | Cancel and go back to main mode |
| <b>FUNCT+6</b>     | Toggle statusbar visibility     |
| <b>FUNCT+8</b>     | Help screen                     |

## Line and box mode:

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Pressing **L** in the main mode starts the Line and box mode. In this mode lines and boxes can be drawn, plotting with the present selected screencode and attribute value.

Ensure that the cursor is located at a corner of the selection to be made before entering Select mode. On entering select mode, grow the selection by pressing the **Cursor keys** to increase or decrease width and height in the desired direction from the origin. Leaving with or height at one character draws a line, otherwise a box is drawn.

**OSE-LOCI new:** while growing the box, press **O** to toggle between a filled box (default, matching V1) and a hollow box that only plots the four border lines, leaving the interior untouched. The toggle can be flipped back and forth as many times as you like before accepting.

**OSE-LOCI new:** also while growing the box, press **C** to toggle between a rectangular box (default) and an ellipse/circle inscribed in the same bounding box. Combine with **O** for a hollow ellipse outline, or leave **O** off for a filled ellipse. Note that since character cells are 6x8 pixels (not square), a square bounding box renders as a flattened ellipse, not a perfect circle – widen the box if you want a rounder result.

Accept with **RETURN**, cancel with **ESC**. Both will leave this mode and return to main mode.

**FUNCT+8** will show a help screen with all screen commands for this mode.

| Key                | Description                                              |
|--------------------|----------------------------------------------------------|
| <b>Cursor keys</b> | Expand/shrink in the selected direction                  |
| <b>O</b>           | <b>OSE-LOCI new:</b> Toggle filled/hollow box or ellipse |
| <b>C</b>           | <b>OSE-LOCI new:</b> Toggle rectangle/ellipse shape      |
| <b>RETURN</b>      | Accept line, box or ellipse                              |
| <b>ESC</b>         | Cancel and go back to main mode                          |
| <b>FUNCT+6</b>     | Toggle statusbar visibility                              |
| <b>FUNCT+8</b>     | Help screen                                              |

## Write mode:

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Pressing **W** in the main mode starts the Write mode. In this mode text can be freely entered by using the full keyboard, making text input way easier than selecting the appropriate screencodes one by one. The full keyboard is supported, as long as the characters entered are printable (screencodes higher than 32).

Colors and attribute can be plotted while in write mode by first modifying the attributes desired, then by plotting an attribute code: - Pressing **CTRL+Z** or **CTRL+X** decreases resp. increases selected ink color - Pressing **\*CTRL+C** or **CTRL+V** decreases resp. increases selected paper color - Pressing **CTRL+B**, **CTRL+A** or **CTRL+D** toggles alternate charset, double resp. blink attribute - Pressing **CTRL+R** toggles reverse mode - Pressing **FUNCT+1** plots ink - Pressing **FUNCT+2** plots paper - Pressing **FUNCT+3** plots character modifier. - **OSE-LOCI new:** Pressing **FUNCT+5** opens a popup to enter an attribute directly as a hex digit: choose **1** for ink, **2** for paper or **3** for character modifier, then type a value 0-7. This is an alternative to cycling with **CTRL+Z/X/C/V** when you already know the exact value you want.

Leave Write mode by pressing **ESC**. **FUNCT+8** will show a help screen with the key commands for this mode.

| Key                | Description                                                  |
|--------------------|--------------------------------------------------------------|
| <b>Cursor keys</b> | Move in the selecOric direction                              |
| <b>DEL</b>         | Clear present cursor position (plot white space)             |
| <b>CTRL+A</b>      | Toggles alternate charset attribute                          |
| <b>CTRL+B</b>      | Toggles blink attribute                                      |
| <b>CTRL+D</b>      | Toggles double attribute                                     |
| <b>CTRL+R</b>      | Toggles reverse                                              |
| <b>CTRL+Z</b>      | Decreases ink color                                          |
| <b>CTRL+X</b>      | increases ink color                                          |
| <b>CTRL+C</b>      | Decreases paper color                                        |
| <b>CTRL+V</b>      | increases paper color                                        |
| <b>FUNCT+1</b>     | Plots ink                                                    |
| <b>FUNCT+2</b>     | Plots paper                                                  |
| <b>FUNCT+3</b>     | Plots character modifier                                     |
| <b>FUNCT+5</b>     | <b>OSE-LOCI new:</b> Enter ink/paper/modifier as a hex digit |
| <b>ESC</b>         | Go back to main mode                                         |
| <b>FUNCT+6</b>     | Toggle statusbar visibility                                  |
| <b>FUNCT+8</b>     | Help screen                                                  |
| <b>Other keys</b>  | Plot corresponding character (if printable)                  |

## Color value reference:

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Below the overview of the 8 color values.

The values are calculated from the 3 bits used for the Red, Green and Blue (RGB) bits: - +1 for Red - +2 for Green - +4 for Blue

| Number | Color   | B-G-R |
|--------|---------|-------|
| 0      | Black   | 0-0-0 |
| 1      | Red     | 0-0-1 |
| 2      | Green   | 0-1-0 |
| 3      | Yellow  | 0-1-1 |
| 4      | Blue    | 1-0-0 |
| 5      | Magenta | 1-0-1 |
| 6      | Cyan    | 1-1-0 |
| 7      | White   | 1-1-1 |

### Serial attribute code reference:

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The Oric does not have a separate attribute memory space, changing any attribute is done by plotting an attribute code where normally a character would go, with the effect of that attribute valid for the rest of the line until another attribute code overrides it in the rest of the line to the right. Also, an attribute code can either change the ink, change the paper, change character set modifiers or change video control attributes, not any combination of these four at the same time. If you want to change two or more of these four categories, you have to plot the same number of attribute code after each other. That is why they are called serial attributes. This rather complicates screen design in multi color as every color change does cost a spot where no normal character can go.

In Oric Screen Editor, all attributes but the video control attributes are supported. But OSE is not aware of attributes you have placed in the line, so proper attribute placement is something you as user should take care of in the design.

Attributes codes are all plot codes from 0 to 31, codes from 32 to 127 are the printable characters according to standard ASCII codes, codes from 128 and up are the same but in reverse video.

See for full background and reference: <https://osdk.org/index.php?page=articles&ref=ART9>

Overview of possible attribute codes:

*Codes 0-7: Change ink*

To change the ink color, the codes are the basic color numbers as mentioned above in the Color value reference, so basically just setting bits 0,1 and 2 for the RGB value, bit 3-7 at zero.

Bitpattern:

| Bit     | 7-6-5 | 4 | 3 | 2    | 1     | 0   |
|---------|-------|---|---|------|-------|-----|
| Meaning | 0     | 0 | 0 | Blue | Green | Red |

| Code | Hex | 0-0-0-P-C-B-G-R | Ink Color |
|------|-----|-----------------|-----------|
| 00   | 00  | 0-0-0-0-0-0-0-0 | Black     |
| 01   | 01  | 0-0-0-0-0-0-0-1 | Red       |
| 02   | 02  | 0-0-0-0-0-0-1-0 | Green     |
| 03   | 03  | 0-0-0-0-0-0-1-1 | Yellow    |
| 04   | 04  | 0-0-0-0-0-1-0-0 | Blue      |
| 05   | 05  | 0-0-0-0-0-1-0-1 | Magenta   |
| 06   | 06  | 0-0-0-0-0-1-1-0 | Cyan      |
| 07   | 07  | 0-0-0-0-0-1-1-1 | White     |

#### *Codes 8-15: Character set modifier*

With bit 3 enabled (so add 8) bits 0,1 and 2 are used to modify charset behavior. The different charset behaviors can be set at one time with one code.

Bitpattern:

| Bit     | 7-6-5 | 4 | 3                   | 2        | 1              | 0            |
|---------|-------|---|---------------------|----------|----------------|--------------|
| Meaning | 0     | 0 | Charset modifier on | Blink on | Double size on | Alternate on |

| Code | Hex | 0-0-0-P-C-B-D-A | Effect on charset                          |
|------|-----|-----------------|--------------------------------------------|
| 08   | 08  | 0-0-0-1-0-0-0-0 | Use standard charset                       |
| 09   | 09  | 0-0-0-1-0-0-0-1 | Use alternate charset                      |
| 10   | 0A  | 0-0-0-1-0-0-1-0 | Use double size standard charset           |
| 11   | 0B  | 0-0-0-1-0-0-1-1 | Use double size alternate charset          |
| 12   | 0C  | 0-0-0-1-0-1-0-0 | Use blinking standard charset              |
| 13   | 0D  | 0-0-0-1-0-1-0-1 | Use blinking alternate charset             |
| 14   | 0E  | 0-0-0-1-0-1-1-0 | Use double size blinking standard charset  |
| 15   | 0F  | 0-0-0-1-0-1-1-1 | Use double size blinking alternate charset |

#### *Codes 16-23: Change paper*

To change the paper color, the codes are the basic color numbers as mentioned above in the Color value reference, so basically just setting bits 0,1 and 2 for the RGB value, together with setting bit 4 (so adding 16). Bit 3, and bits 5,6 and 7 should be 0.

Bitpattern:



| Bit     | 7-6-5 | 4               | 3 | 2    | 1     | 0   |
|---------|-------|-----------------|---|------|-------|-----|
| Meaning | 0     | Paper modify on | 0 | Blue | Green | Red |

| Code | Hex | 0-0-0-P-C-B-G-R | Paper Color |
|------|-----|-----------------|-------------|
| 16   | 10  | 0-0-0-1-0-0-0-0 | Black       |
| 17   | 11  | 0-0-0-1-0-0-0-1 | Red         |
| 18   | 12  | 0-0-0-1-0-0-1-0 | Green       |
| 19   | 13  | 0-0-0-1-0-0-1-1 | Yellow      |
| 20   | 14  | 0-0-0-1-0-1-0-0 | Blue        |
| 21   | 15  | 0-0-0-1-0-1-0-1 | Magenta     |
| 22   | 16  | 0-0-0-1-0-1-1-0 | Cyan        |
| 23   | 17  | 0-0-0-1-0-1-1-1 | White       |

Note that in OSE calculation of these attribute codes by yourselves is not necessary, the program will do so for you given the selected attributes and color. In memory however this is how the codes are stored.

## File format reference

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As the Oric does not have separate attribute memory, screendata is basically just a *widthheight dump of screen codes*. *The screen file is a flat data file with these screencodes, length is calculated as widthheight*. So a standard 40x27 screen would be 1.080 bytes.

## Credits

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Oric Screen Editor

Screen editor for the Oric Atmos

Written in 2022 by Xander Mol

Based on VDC Screen Editor for the C128

<https://github.com/xahmol/OricScreenEditor>

<https://www.idreamtin8bits.com/>

Code and resources from others used:

- CC65 cross compiler:  
<https://cc65.github.io/>

- 6502.org: Practical Memory Move Routines: Starting point for memory move routines  
[http://6502.org/source/general/memory\\_\\_move.html](http://6502.org/source/general/memory__move.html)
- DraBrowse source code for DOS Command and text input routine  
DraBrowse (db\*) is a simple file browser. Originally created 2009 by Sascha Bader. Used version adapted by Dirk Jagdmann (doj)  
<https://github.com/doj/dracopy>
- lib-sedoric from oricOpenLibrary (for SEDORIC file operations By Raxiss, (c) 2021 <https://github.com/iss000/oricOpenLibrary/blob/main/lib-sedoric/libsedoric.s>
- Bart van Leeuwen and [forum.defence-force.org](http://forum.defence-force.org): For inspiration and advice while coding.
- jab / Artline Designs (Jaakko Luoto) for inspiration for Palette mode and PETSCII visual mode
- Original windowing system code on Commodore 128 by unknown author.
- Tested using real hardware Oric Atmos plus Cumana Reborn, and Oricutron for Windows and Linux

The code can be used freely as long as you retain a notice describing original source and author.

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